Tutorial 4

1. Experience has shown that while walking in a certain park, the time X, in minutes, between seeing two people running has a density function of the form

$$f(x) = \begin{cases} \lambda x e^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

   (a) Calculate the value of λ.

   (b) Find the CDF of X.

   (c) What is the probability that Jeff, who has just seen a person running, will see another person running in 2 to 5 minutes? In at least 7 minutes?

2. Sketch the graph of the function

$$f(x) = \begin{cases} \frac{1}{2} - \frac{1}{4}|x-3|, & 1 \le x \le 5, \\ 0, & \text{otherwise,} \end{cases}$$

   and show that it is the probability density function of a random variable X.

   (b) Find F, the distribution function of X, and show that it is continuous.

   (c) Sketch the graph of F.

3. The lifetime of a TV tube (in years) is an exponential random variable with $\lambda = 0.1$. If Jim bought his TV set 10 years ago, what is the probability that its tube will last another 10 years?

4. The scores on a test given to 100,000 students are normally distributed with

$$\mu = 500, \qquad \sigma = 100.$$

   What should the score of a student be to place him among the top 10% of all students?

5. Starting at 5:00 A.M., every half hour there is a flight from San Francisco airport to Los Angeles International airport. Suppose that none of these planes is completely sold out and that they always have room for passengers. A person who wants to fly to L.A. arrives at the airport at a random time between 8:45 A.M. and 9:45 A.M. Find the probability that she waits

   (a) at most 10 minutes;

   (b) at least 15 minutes.